



EAST WEST UNIVERSITY

Department of Mathematical and Physical Sciences(MPS)

Quiz-1

Fall-2025

PHY-109 : Engineering Physics - 1

Course Instructor : Aswariya Roy

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Time: 25 min

Marks: 10

Section: 5

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**Question 1:** An elevator cab of mass  $m = 400$  kg is descending with an initial speed  $v_i = 5.0$  m/s. Suddenly, the supporting cable begins to slip, allowing the cab to accelerate downward with a constant acceleration  $a = g/3$ . The cab falls a distance  $d = 12$  m.

- (a) Calculate the work done by gravity,  $W_g$  and the tension in the cable,  $W_T$  during this  $d = 12$  m fall. [3]
- (b) If A small resistive force begins to act upward, then calculate the work done by the resistive force. Also determine the net work  $W_{\text{net}}$  on the cab and final speed  $v_f$  of the cab considering the resistive force. [3.5]

**Question 2:** A 3 kg ball drops vertically onto a floor, hitting it with a speed of 35 m/s. It rebounds with a speed of 20 m/s. What impulse acts on the ball during the contact? [1.5]

**Question 3:** Give Statement of Work-Energy theorem with equation.[2]



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Quiz-2

Fall-2025

PHY-109 : Engineering Physics - 1

Course Instructor : Aswariya Roy

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Time: 30 min

Marks: 10

Section: 5

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**Question 1:** Water flows through a horizontal pipe. The water flow rate in a pipe is  $0.02 \text{ m}^3/\text{s}$ .

- (a) If the pipe's diameter at one point is 10 cm, calculate the velocity of water at that point. [1.5]
- (b) If the pipe narrows to a diameter of 5 cm, find the velocity in the narrower section. [2]
- (c) Calculate pressure difference between the wide and narrow sections. [2]

**Question 3:** Give Statement and write down Bernoulli's Equation, also identify each term. [2]

**Question 4:** A solid disc of mass  $M = 4.0 \text{ kg}$  and radius  $R = 0.25 \text{ m}$  is free to rotate about a fixed horizontal axis passing through its centre. A light string is wound around the rim of the disc, and a hanging mass  $m = 1.0 \text{ kg}$  is attached to the free end of the string. When the mass is released from rest, it falls vertically, causing the disc to rotate. Neglect air resistance and assume that the string does not slip on the surface of the disc. The tension of the string is 25 N.

- (a) Draw freebody diagram for this problem and by writing the equation of motion for the mass  $m$ , find it's acceleration. [2.5]

of water in the narrower section. [2]

**Question 2:** Water flows out of a large open tank through a small hole at the bottom. If the water level is  $h = 5\text{ m}$  above the hole, find the velocity of the water exiting the hole. Assume negligible velocity at the tank surface and  $\rho = 1000\text{ kg/m}^3$ . [3]

**Question 3:** Give Statement and write down Bernoulli's Equation, also identify each term. [2.5]

**Question 4:** A solid disc of mass  $M = 4.0\text{ kg}$  and radius  $R = 0.25\text{ m}$  is free to rotate about a fixed horizontal axis passing through its centre. A light string is wound around the rim of the disc, and a hanging mass  $m = 10\text{ kg}$  is attached to the free end of the string. When the mass is released from rest, it falls vertically with acceleration  $a = 15.2\text{ m/s}^2$ , causing the disc to rotate. Neglect air resistance and assume that the string does not slip on the surface of the disc.

- (a) Draw freebody diagram for this problem and by writing the equation of motion for the mass  $m$ , find the tension of the string. [2.5]